

Blender comes with a number of important tools, the most essential being its in-built editor. Editing operation can be carried out in two modes: granular-level Edit mode or macro-level Object mode. When editing a design mesh, Edit mode allows you to customise each vertex of the mesh individually. On the other hand, Object mode lets you modify multiple objects together as a unit. Some actions that can be taken in Object mode editing are moving, scaling or rotation of entire polygon meshes.

Blender projects can consist of multiple data blocks related to the project in the form of objects, meshes, scenes, material or lamps. Materials define the appearance of meshes and geometric objects, and consist of three shaders: surface, volume and displacement. Blender also allows different simulations like fluid simulation, cloth simulation effects and so on.

It has a dedicated video editing tool called Video Sequence Editor (VSE). It is fully functional with useful features like video transformations, blur effects, colour fading, transition effects and more.

The software can open separate windows within the platform for separate operations. The interface is highly customisable, be it in terms of screen size, usability or workspace management. For example, you can dedicate the interface to a specific task (such as video editing, texturing or UV mapping), keeping unrelated features hidden.

Blender saves its projects as .blend files. These files are backwards and forwards compatible with most versions of Blender. It also

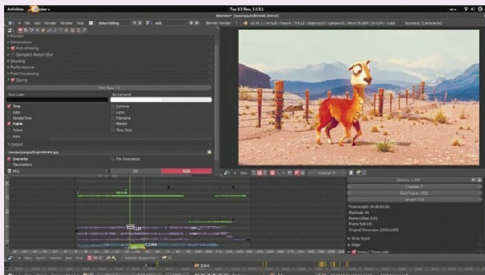


Fig. 3: Video editing in Blender (Credit: Blender.org)

retains the customised settings of the UI while being created. Blender has import and export properties, too. Various 3D file formats like 3DS, FBX, DXF, SVG, X3D, STL and VRML are supported.

Blender files can be run on Web browsers using Blend4Web, which is an open source WebGL framework. It converts Blender file graphics, sound, animation and simulations into browser-compatible formats. You can add modifications of your own to the tool. For example, you can create custom nodes using Open Shading Language.

Blender's latest iteration is 2.79. It comes with a host of new additions, including Denoiser and PBR Shader features. It also brings Shadow Catcher suite. An improved film colour management system has been added, too. In addition, there are more than 700 other bug fixes.

Using Blender

Blender helps animate the thoughts of the creative mind. It can be used across applications like creating ba-

Recommended system specifications

These are the recommended system specifications for running Blender efficiently:

- 64-bit quad-core unit processor (minimum 32-bit dual-core)
- Preferably 8GB RAM (minimum 2GB)
- 2GB graphics card (minimum 512MB)
- 1080p full-HD display

sic 3D models, developing videos or even for full-fledged movie and video game production. An active community support makes it more accessible and actionable. This makes it a great tool for beginners as well as for professionals.

Blender has been adopted in several professional use cases till date, some of the most mentionable being the application in 2004 movie *Spiderman 2* and in NASA for developing an interactive Web-interface for Mars Curiosity Rover. The capacity of usage that this software brings with it is tremendous. We recommend you to start using it as well! **EFY**

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